

Here are the essential features:

1. Certificate Lifecycle Management

- **Certificate creation, renewal, and revocation:** The platform should support comprehensive management of the certificate lifecycle (issuance, renewal, revocation).
- **Qualified certificates and international standards (eIDAS):** Ensure legal recognition of electronic signatures through compliance with qualified certificate standards.
- **Chain of trust:** Use a hierarchical infrastructure (Root CA and Intermediate CAs) to establish a solid certification chain.
- **Number of Digital Certificate:** We will need a capacity of 1 million expandable.
- **CRL service:** List of certificates that have been revoked and are no longer trusted.
- **OCSP service:** Real-time verification of a certificate's validity without downloading an entire Certificate Revocation List (CRL).
- **Recent and old encryption/hash algorithms:** Ability to use a recent or old algorithm/hash to issue certificates.
- **Different formats:** Issue certificate in different formats (.pem, .crt, .cer, .p12, .pfx, .p7b/.p7c)

2. Electronic Signature Features

- **Qualified electronic signatures:** Compliance with local and international regulations, with protection against forgery.
- **Signature customization:** Ability to add text, images, and QR codes to signatures.
- **Secure timestamping:** Integration of timestamping to guarantee the temporal validity of signatures.
- **Multi-signature management:** Capability to sign documents with multiple signatories or to request their signature

3. Security and Compliance

- **Private key protection:** Hardware security modules (HSM) for secure storage of private keys.
- **Strong authentication:** User authentication via tokens, secure passwords, or certificates.

Action tracking: Audit and traceability of signatures and transactions.

Profile management and authorization: User profile management.

- **CAs management:** Ability to import and export CAs.

- **Several languages:** English and French language
- **Physical appliance:** Our policies advise us to have a physical appliance in our country.
- **Fast queue management:** Ability to issue several certificates per minute (like 200 or more).

4. Accessibility and Integration

- **Integration API:** API to embed signing functionalities into third-party applications (workflows, management platforms). Detailed documentation to connect the PKI platform with our existing solutions.
- **User-friendly interface:** A web interface for managing signatures and certificates.
- **Secure storage and archiving:** Preservation of electronic signature evidence and certificates to maintain their legal validity over time.

5. Continuity and Support

- **Disaster Recovery Plan (DRP):** Ensure continuity in case of incidents, with fast access to critical certificates and keys.
- **Technical support:** Assistance to resolve issues promptly and maintain high service availability.
- **Help with installation and configuration:** We will need your support for the first installations and configurations of the platform.

6. Statistics and Reporting

- **Performance tracking:** Reporting on signature volumes, certificate statuses, and resource utilization.
- **Detailed logs:** we need a detailed log to improve our services. Time, date, user logged in, action performed.